

Palmistry & Tarot Intelligence Platform

Milestone : Milestone 1

To achieve the objectives of Milestone 1, the following activities were carried out:

1. Project Understanding

- Studied the project requirements and objectives of the **Palmistry & Tarot Intelligence Platform**.
- Identified the required datasets and supporting technologies such as Google Colab, Python, Pandas, Matplotlib, and MediaPipe.

2. Dataset Collection

- Downloaded the **Hands and Palm Images Dataset** from Kaggle.
- Downloaded the **Tarot JSON Dataset** from Kaggle.
- Collected the **MediaPipe Hands** framework for future hand landmark detection.

3. Google Colab Environment Setup

- Connected Google Drive with Google Colab.
- Uploaded both datasets to Google Drive.
- Extracted the compressed dataset files within the Colab environment.

```
from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive

ls "/content/drive/MyDrive/palmistry_tarot_project"
hands-and-palm-images-dataset.zip  tarot-json.zip

!unzip "/content/drive/MyDrive/palmistry_tarot_project/tarot-json.zip" -d "/content/tarot_dataset"

Archive: /content/drive/MyDrive/palmistry_tarot_project/tarot-json.zip
  inflating: /content/tarot_dataset/cards/c01.jpg
  inflating: /content/tarot_dataset/cards/c02.jpg
```

4. Tarot Dataset Analysis

- Loaded the JSON dataset using Python.
- Converted the JSON data into a Pandas DataFrame.
- Performed Exploratory Data Analysis (EDA), including:
 - Dataset structure analysis
 - Shape and feature inspection
 - Missing value analysis
 - Duplicate record checking

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 78 entries, 0 to 77
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   name                   78 non-null    object  
1   number                 78 non-null    object  
2   arcana                 78 non-null    object  
3   suit                  78 non-null    object  
4   img                   78 non-null    object  
5   fortune_telling        78 non-null    object  
6   keywords               78 non-null    object  
7   meanings               78 non-null    object  
8   Archetype              22 non-null    object  
9   Hebrew Alphabet       22 non-null    object  
10  Numerology             62 non-null    object  
11  Elemental              38 non-null    object  
12  Mythical/Spiritual     22 non-null    object  
13  Questions to Ask       78 non-null    object  
14  Astrology              40 non-null    object  
15  Affirmation            56 non-null    object  
dtypes: object(16)
memory usage: 9.9+ KB
```

Observation

- Total Tarot Cards: **78**
- Total Features: **16**
- No duplicate records.
- Missing values were present only in optional columns.

5. Palm Image Dataset Analysis

- Explored the folder structure of the palm image dataset.
- Counted the total number of images.
- Displayed sample palm images.
- Verified image dimensions and file format.
- Checked the dataset for corrupted images.

```
import os

image_folder = "/content/palm_dataset/Hands/Hands"

files = os.listdir(image_folder)

print("Total files:", len(files))
print("\nFirst 10 files:")

for file in files[:10]:
    print(file)

Total files: 11076

First 10 files:
Hand_0001179.jpg
Hand_0011418.jpg
Hand_0010476.jpg
Hand_0001170.jpg
Hand_0007864.jpg
Hand_0009112.jpg
Hand_0010704.jpg
Hand_0007116.jpg
Hand_0007480.jpg
Hand_0000637.jpg
```

```

from PIL import Image
import os

corrupted = []

for file in os.listdir(image_folder):
    path = os.path.join(image_folder, file)
    try:
        img = Image.open(path)
        img.verify()
    except Exception:
        corrupted.append(file)

print("Corrupted Images:", len(corrupted))

Corrupted Images: 0

```

Observation

- Total Images: **11,076**
- Resolution: **1600 × 1200**
- Image Format: JPG
- Corrupted Images: **0**

6. MediaPipe Setup

- Installed the MediaPipe library.
- Verified the installation by checking the installed version.
- Prepared the environment for future hand landmark detection.

```

import mediapipe as mp

print("MediaPipe Version:", mp.__version__)

MediaPipe Version: 0.10.35

```

7. Outcomes of Milestone1

- Both datasets were successfully uploaded and analyzed.
- Data quality was verified through EDA.
- The Google Colab environment was configured successfully.
- MediaPipe was installed and verified for future development stages.

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